SDG Blockchain Accelerator Cohort 2

Roadmap

Accelerator Duration: 29 September 2025 - 19 January 2026
Challenge Owner (CO): UNDP Nigeria
Solution Maker (SM): CUBID Protocol

1. Challenge Definition

1.1 Refined Challenge Statement

Describe the real-world development challenge your solution addresses.

Rural communities across Nigeria remain underserved by the national electricity grid and rely heavily on expensive, unreliable, and polluting energy sources such as diesel and gasoline generators and kerosene. While solar mini-grids are a proven technical solution, their long-term sustainability is constrained by weak investment models, tariff collection systems, fragmented financial records, and limited transparency.

Mini-grid operators struggle to demonstrate predictable revenues, governments and development partners lack reliable data for oversight, and investors face high verification costs. As a result, viable solar infrastructure often fails to scale, carbon finance remains inaccessible, and rural households continue to face high energy costs and unreliable service. The core challenge is therefore not generation capacity, but **the absence of trusted, transparent, and inclusive operating and reporting infrastructure for rural solar mini-grids**.

1.2 Local Context

Why does this challenge matter? Describe the setting and key beneficiaries. Explain why this challenge is important in your specific country or regional context. Describe the affected communities or sectors, any existing efforts to solve the issue, and the roles of key stakeholders involved.

Nigeria has one of the largest electricity access gaps globally, with rural and peri-urban communities disproportionately affected. In many farming regions, the lack of reliable power

limits irrigation, milling, cold storage, healthcare delivery, and local enterprise development. Solar mini-grids are increasingly deployed through public, private, and donor-funded programs, yet many struggle after installation due to poor revenue collection, manual reporting, and weak accountability mechanisms.

Key beneficiaries include rural households, agricultural producers, community cooperatives, and local mini-grid operators. Existing efforts by the Rural Electrification Agency (REA), the African Minigrid Program (AMP), and donor-supported initiatives have expanded infrastructure, but **system-level operating data and financial transparency remain fragmented**. See section 3.5 Market Size below for more quantified details on impact.

This challenge matters acutely in Nigeria because it directly affects energy access, climate mitigation, rural livelihoods, and the credibility of emerging carbon markets. Without trusted records of electricity production and consumption, carbon credits cannot be efficiently verified, limiting access to climate finance and undermining long-term sustainability.

1.3 Alignment with UNDP Country Office and National Development Priorities

Explain how the proposed solution aligns with the strategic objectives of the UNDP Country Office and national development frameworks. Does it support existing programs or government priorities (e.g., digital transformation, climate resilience, youth employment, financial inclusion)? Highlight any synergies with UNDP initiatives, government strategies, or donor-funded efforts.

The Solar Village pilot aligns strongly with UNDP Nigeria's priorities in **energy access, climate action, digital public infrastructure, financial inclusion, and systems-level innovation**. It supports national objectives under Nigeria's Energy Transition Plan, rural electrification strategies led by REA, and the emerging Nigerian carbon market framework coordinated by the National Council on Climate Change (NCCC).

The solution advances:

- **Digital transformation** by introducing verifiable, interoperable data infrastructure for energy systems
- **Climate resilience and mitigation** through verifiable emissions reduction and carbon credit readiness
- **Financial inclusion** by enabling prepaid electricity via familiar payment channels
- **Institutional transparency** by improving auditability for public and donor-funded assets

The pilot complements existing UNDP-supported electrification initiatives by strengthening operational viability rather than duplicating infrastructure deployment. In our initial pilot we have identified **23 villages with minigrids recently developed or currently underway**, each with on average 20 households, for a total of 200+ households potentially impacted in Phase 1. Some benefits Solar Village will bring to these villages include:

- Most of these minigrids don't have an integrated payment solution. Solar Village will provide native options to collect payments. Notably, we will enable payment by Agent - where one or more villagers with access to banking will pay-on-behalf-of those who are unbanked.
- All of these minigrids are obliged to report their numbers to REA on a monthly basis, which they currently submit manually. Solar Village will automate this reporting and make it available on a daily basis. Our reporting will be aligned with the AMP QAMF standards, which makes it immediately possible to include the reporting into pan-African solar database without any manual involvement.
- None of these microgrids are producing carbon credits, or at least they are not producing carbon credits in alignment with recognized frameworks / made available on recognized marketplaces. Solar Village will add this benefit to all 23 pilot minigrids, which will contribute to cover their expenses and keep electricity pricing low.

1.4 Relevance to UNDP Programmes and Resource Mapping

(Is the pilot building on an existing initiative, or is it a new standalone project? If the project builds on an existing initiative, provide further details. Include scope, scale, timeline and available co-financing or in-kind contributions.

The pilot builds on existing rural electrification efforts rather than operating as a standalone intervention. Phase 2 focuses on operational infrastructure for selected solar mini-grids, leveraging prior capital investments in generation assets.

Scope & scale:

- Initial deployment with selected mini-grids serving rural communities
- Meter-level data collection at 15-minute intervals
- Shared dashboards for operators and UNDP teams

Timeline:

- Phase 2 pilot: operational validation and reporting
- Designed for replication across Nigeria

Resources:

- In-kind contributions from mini-grid operators (existing infrastructure)
- Technical development by the Solution Maker
- Oversight and coordination by UNDP Nigeria

1.5 Key Stakeholders & Potential Partners

(Please list all the partners involved in this project. List all relevant parties who will interact with, benefit from, or influence the solution (this may include government agencies, NGOs, community members, or tech partners.))

The following parties will interact with, benefit from, or influence the solution

- **UNDP Nigeria** – programme oversight, monitoring, and learning
- **Rural Electrification Agency (REA)** – sector alignment and scale pathway
- **National Council on Climate Change (NCCC)** – carbon market alignment
- **African Minigrid Program (AMP)** – implementation ecosystem
- **Solar mini-grid operators** – system users and beneficiaries
- **Rural households and community cooperatives** – end users
- **Cubid Protocol** – digital identity and tamper-evident transaction infrastructure
- **Local financial institutions and payment providers** – payment rails
- **Carbon standards bodies (e.g. Gold Standard, Verra)** – future verification partners

1.6 Expected Impact

(Outline intended outcomes from the CO's perspective. What would a successful pilot enable - policy change, improved services, empowerment, systems improvement, or scaling innovation?)

From the Country Office perspective, a successful pilot would:

- Improve transparency and accountability in rural energy operations
- Enable verifiable reporting of electricity generation and consumption
- Lower the cost and friction of carbon credit verification
- Strengthen the financial sustainability of mini-grid operators
- Create a scalable, standards-aligned model for national replication

Critically, it would demonstrate how **digital public infrastructure can convert existing energy assets into measurable, finance-ready systems**, unlocking new policy and funding pathways.

1.7 Target SDGs and Indicators

(List up to three specific SDG indicators your challenge and proposed solution will directly contribute to. Focus on indicators you will actively address, not indirect ones.)

The pilot directly contributes to the following SDGs and indicators:

- **SDG 7 – Affordable and Clean Energy**
  Indicator 7.1.1: Proportion of population with access to electricity

- **SDG 9 – Industry, Innovation and Infrastructure**
  Indicator 9.1.1: Proportion of rural population with access to all-season infrastructure services

- **SDG 13 – Climate Action**
  Indicator 13.2.2: Total greenhouse gas emissions per year, supported by verified emissions reductions from solar mini-grids

2. User Mapping & Testing Strategy

This section outlines how users and stakeholders were identified, engaged, and incorporated into the design and validation of the solution. It describes the primary user personas, the testing approach used during development, and the feedback mechanisms applied to ensure the solution responds to real operational needs and constraints in the pilot context.

2.1 Primary User Persona

Describe the main user or beneficiary of your solution.

Primary user: Rural household electricity consumers connected to a solar mini-grid

The primary beneficiary is a rural household in Nigeria that relies on a community solar mini-grid for electricity. This user typically has limited access to formal banking infrastructure, may not own a smartphone, and is highly sensitive to electricity cost, reliability, and ease of payment. Electricity is used for lighting, phone charging, household appliances, and increasingly for productive activities such as irrigation, milling, refrigeration, and small enterprises. Solar Village not only makes electricity more accessible, but it also makes it more affordable (see Carbon Credits below.

Primary customer: Solar mini-grid operators

The primary customer of Solar Village is the solar mini-grid operator responsible for deploying, operating, and maintaining community energy systems. These operators face tight margins, high upfront capital costs, and ongoing pressure to demonstrate financial viability to investors, regulators, and donors. Revenue collection is often fragmented, reporting is manual, and proving performance over time is costly and error-prone. Solar Village addresses these challenges by providing operators with a transparent, automated operating layer that links metered electricity production and consumption to verifiable financial and performance records. By capturing energy data at the source and anchoring it immutably, operators gain credible evidence of system performance, enabling them to **mint and sell carbon credits based on verified kWh generation and diesel displacement** once aligned with applicable standards. The resulting carbon revenues can supplement tariffs, improve unit economics, and either lower consumer prices or increase system sustainability, making the platform financially attractive while strengthening long-term operational resilience.

Secondary users include: Investors, payment collectors (agents), government agencies, and UNDP programme teams, but the system is explicitly designed around the needs and constraints of the **end household**, ensuring inclusion regardless of device type, literacy level, or connectivity.

2.2 User Story

Use this format: "As a [user], I want to [goal], so that [value]."

- **As a rural household**, I want to easily purchase electricity using familiar payment methods, **so that** I can reliably access power without traveling long distances or relying on intermediaries.

- **As a mini-grid operator**, I want transparent, automated records of electricity consumption and payments, **so that** I can operate sustainably and demonstrate financial viability.

- **As an agent (payment collector),** I want a simple and transparent way to collect electricity payments and confirm meter activation, **so that I** can support households in my community, reduce cash handling, ensure payments are accurately recorded without disputes, and make some money doing this.

- **As an investor or donor,** including **UNDP and government stakeholders,** I want verifiable, auditable records of electricity production, consumption, and repayment flows, **so that** I can confidently finance solar mini-grids, track impact over time, ensure accountability, reduce the risk of misreporting or double counting, and inform policy and scale-up decisions.

2.3 Testing Strategy

How and when will user testing be conducted?

User testing will be conducted **in phased, real-world conditions** during the pilot deployment.

- **Pre-deployment testing:**
    - Usability testing of the mobile web and USSD flows with a small group of households and operators
    - Validation of language, menu structure, and transaction clarity
    - Dry runs of meter activation and payment confirmation

- **Live pilot testing:**
    - Testing occurs during real electricity purchases and daily usage
    - Monitoring of payment success rates, activation latency, and error cases
    - Comparison of smartphone vs feature phone user experience

- **Ongoing testing:**
    - Continuous observation during the pilot period

    - Edge-case testing during outages, delayed connectivity, or partial payments

Testing is intentionally embedded into **actual operational use**, rather than simulated lab environments, to ensure relevance to real constraints.

2.4 Feedback Loop Design

How will feedback from users and stakeholders be gathered and integrated?

Feedback is gathered and integrated through **multiple, complementary channels**:

- **Direct user feedback:**
    - Short in-person interviews and community check-ins during pilot rollout
    - Simple satisfaction questions via USSD or assisted sessions
- **Operator feedback:**
    - Structured feedback sessions with mini-grid operators on payment flows, reporting clarity, and operational impact
    - Review of reconciliation issues, tariff logic, and support requests
- **System-level feedback:**
    - Analysis of usage data, failed transactions, delayed activations, and drop-offs
    - Dashboard reviews with UNDP teams to assess reporting adequacy and decision usefulness
- **Iteration mechanism:**
    - Feedback is reviewed in regular pilot retrospectives
    - Minor usability improvements are released continuously
    - Structural changes are logged and incorporated into subsequent rollout phases

This feedback loop ensures the platform evolves based on **observed behavior and stakeholder needs**, not assumptions, and supports learning that can inform future UNDP programming and national scale-up.

3. Solution Overview

This section provides an overview of the Solar Village solution, describing its core purpose, key features, and underlying technology. It explains how blockchain-enabled infrastructure is used to improve transparency, accountability, and sustainability in the operation of rural solar mini-grids, while remaining accessible and practical for real-world deployment.

3.1 Solution Summary

Briefly describe your proposed blockchain-based solution.

Solar Village is a blockchain-enabled operations platform for rural solar mini-grids. It creates **tamper-evident energy and payment records** by recording metering data at regular intervals in a per-mini-grid ledger (Cardano Hydra), then anchoring daily proofs to Cardano mainnet to create a permanent audit trail. This makes electricity delivery and repayments transparent for operators and UNDP oversight, and lays the groundwork for **credible carbon credit verification and issuance** based on verifiable kWh production and consumption.

3.2 Core Features and Functionalities

What are the most important features of your solution?

Energy data & reporting

- Automated capture of production/consumption data at frequent intervals (e.g., every 15 minutes) to turn each kWh into a verifiable record.
- Role-based dashboards for operators and oversight stakeholders (UNDP, regulators, donors/investors) to monitor performance and outcomes.

Payments & last-mile access

- Prepaid electricity top-ups through familiar local payment channels, designed to work for both smartphone and feature-phone users via web and USSD flows.
- Meter activation tied to confirmed payment, using either remote activation (online meters) or time-bound codes (offline fallback) to ensure inclusion in low-connectivity environments.

Carbon readiness

- A standards-aligned foundation for carbon credit issuance by anchoring real energy data first, enabling lower-cost verification and reduced fraud vectors (e.g., double counting).

Ecosystem extensibility

- Open APIs to allow other mini-grid operators to join and to support replication beyond the pilot footprint.

3.3 Tech Stack and Tools

Which technologies and platforms will you use?

Blockchain back end: Cardano + Hydra (one lightweight Hydra head per mini-grid; daily proofs anchored to Cardano mainnet).

Smart contracts / token layer (as applicable): Cardano native assets and smart-contract settlement patterns for energy/accounting flows and future carbon issuance readiness.

Front end: Mobile-first web application plus USSD interface to support feature phones (via Africa's Talking APIs).

Payments & settlement: Integrations with local banks/mobile money; conversion into **Compliant Naira (cNGN)** as a regulated settlement layer where applicable in the pilot design.

Dashboards & M&E: Real-time analytics views for oversight and reporting

3.4 Blockchain Functionality

What blockchain-specific interactions are included (e.g., token minting, smart contracts, on-chain verification)?

Blockchain-specific interactions include:

- **On-chain logging / anchoring:** Each mini-grid records energy production/consumption in its Hydra head and commits a daily cryptographic proof to Cardano mainnet for tamper-evidence and auditability.

- **On-chain payment traceability (where used):** Electricity top-ups can be logged transparently on-chain via the settlement layer (cNGN), enabling audit-ready visibility into household payments and program fund flows.

- **Tokenization readiness:** The design supports **energy settlement / energy-credit representation** and a pathway to **carbon-credit issuance** (minting) once verified energy data is available and aligned to Gold Standard / Verra / emerging Nigerian standards.

3.5 Market Size

Estimated TAM, SAM, and SOM for Nigeria and then for the whole of Africa.

1. Nigeria Market Estimates

Basis for Calculation:

- **Unit Revenue:** Estimated annual gross revenue per mini-grid is **$30,000**. This is derived from an average pilot size of ~300-500 households/connections, paying affordable solar tariffs significantly lower than the current average diesel cost of $0.52/kWh.
- **Operating Fee:** 5% of Gross Revenue (hypothesis).

Market Stage	Definition	Mini-Grid Count	Gross Revenue ($)	Net Revenue (5%)	Source Rationale
TAM (Total Addressable Market)	**Total Generator Displacement.** Nigeria's 87 million underserved citizens currently spend **$14 billion annually** on expensive, polluting off-grid power (generators). SolarVillage aims to capture and digitize this spend.	**~200,000** (Theoretical max to cover 87M people)	**$14 Billion**	**$700 Million**	87M underserved Nigerians spend $14B/year on generators. This represents the total value flow the platform could theoretically process.
SAM (Serviceable Available Market)	**National Renewable Pipeline.** This covers the immediate pipeline of projects identified by the REA and AMP, including 120 pipeline sites and the "Mission 300" push to electrify 300M Africans (with Nigeria as a major hub).	**~3,000** (Mid-term national scale target)	**$90 Million**	**$4.5 Million**	Based on the goal to reach "national scale" powering tens of thousands of households in Phase 3 and the broader REA pipeline.

SOM (Serviceable Obtainable Market)	**Committed Pilot & Phase 1.** The specifically identified pilot sites confirmed in the AMP and SolarVillage design documents for immediate onboarding.	**23**	**~$690,000**	**~$34,500**	SolarVillage has explicitly secured a pilot for **23 microgrids** in partnership with UNDP/REA.

2. Pan-African Market Estimates

Basis for Calculation:

- **Scaling Factor:** Nigeria represents approximately 15-20% of the Sub-Saharan energy access deficit. The "Mission 300" initiative targets 300 million people across Africa.
- **Unit Revenue:** Maintained at **$30,000** per grid for conservative estimation across diverse economies.

Market Stage	Definition	Mini-Grid Count	Gross Revenue ($)	Net Revenue (5%)	Source Rationale
TAM (Total Addressable Market)	**Universal Access Gap.** 660 million people are projected to lack access in 2030, with 85% in Sub-Saharan Africa. OGS is the least-cost solution for 398 million of them.	**~1 Million** (To cover 398M people)	**$30+ Billion**	**$1.5+ Billion**	Based on the "Universal Access" scenario requiring **$21 billion** in investment for OGS to reach 398 million people.
SAM (Serviceable Available Market)	**Mission 300 & Regional Expansion.** Targets within the World Bank/AfDB "Mission 300" initiative and neighboring West African states (Ghana, etc.) identified in the roadmap.	**~20,000**	**$600 Million**	**$30 Million**	Aligns with the "Mission 300" pledge to electrify 300 million people and SolarVillage's roadmap for cross-border expansion.
SOM (Serviceable Obtainable Market)	**Networked Operators.** Expansion to "thousands of off-grid and mini-grid operators" via the open SaaS model post-pilot.	**~1,000**	**$30 Million**	**$1.5 Million**	SolarVillage aims to become an open-source SaaS platform for operators across Africa after the Nigerian pilot.

Summary of Opportunity

- **Immediate Capture (Nigeria SOM):** The project has a secured, "captive customer" base of **23 mini-grids** generating an estimated **$34,500** in annual net revenue fees immediately upon full deployment.
- **Growth Potential (Nigeria TAM):** The platform sits on a **$14 billion** annual flow of funds currently wasted on diesel generators. Converting even a fraction of this "bad spend" to solar payments represents a massive revenue opportunity.
- **Scale Strategy:** The project is designed to transition from a pilot of 23 sites to a SaaS model accessible to **thousands of operators** across Africa, leveraging the **Mission 300** momentum.

4. Weekly Workplan

Briefly describe your team's focus during each stage of the program. (You may use the workplan template from your hackathon roadmap.)

Over the past three months the team have had standing meetings every Tuesday and Thursday, always with the four core team members (Osondu, Noak, Temitope, and Lantana). Oftentimes we invited guests, and have seen participation from Emurgo Labs, M3tering Protocol, NCCC, REA, alpha test users, UNDP Project leadership, and more. In addition, we also held frequent meetings as Solution Makers to design and build the blockchain features and functionality.

Timeline (Week)	Focus Area	What We Did
1	Challenge Definition	Problem statement and user needs validated with at 3 core stakeholder / team member interviews
2	Scope & Development Approach	Whiteboarded first iteration of the user-facing app.
3	Blockchain Interaction	Drafted overall technical and blockchain interaction architecture.
4	User Interface	Designed and built first version of prototype of mobile-first web app with top-up flow mockup: User Dashboard.
5	USSD Integration	Designed and built second version of web app: Investor Dashboard
6	Data & Dashboard	Designed and built third feature of web app: Minigrid Operator Dashboard, capturing sample transactions to monitoring dashboard
7	Stakeholder Testing –	Designed and built fourth feature of web app:

	Users	Payment Agent Dashboard with payment management features.
8	Carbon Credits	Designed and built fifth feature of web app: Our own SaaS Operator Dashboard, and added management + reporting features for carbon credits.
9	Cardano Hydra finalization + Catalyst application	Finalized the Hydra stack, ironed out bugs, and recorded 50 min video to demonstrate. Wrote and submitted application to Catalyst Fund 15
10	Stakeholder Testing – Donors & Regulators	Demo dashboards shared with potential future user base for feedback
11	Demo Readiness	Finalization of documentation, demo
12	Pilot Preparation + Cardano Accelerator Program Application	Pilot-ready MVP packaged with documentation, training materials, and governance outline. Application submitted to the Cardano Accelerator Program (CAP).

5. Final Deliverables

5.1 Solution Demo (PoC)

A working or visual prototype that demonstrates the core idea. (To be scheduled and presented virtually to CO team and UNDP/Partner FPs in a weekly coordination meeting)

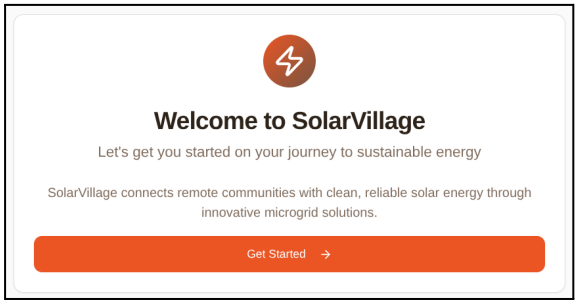
We have developed a visual prototype which was presented to CO team on December 9th, including the following components:

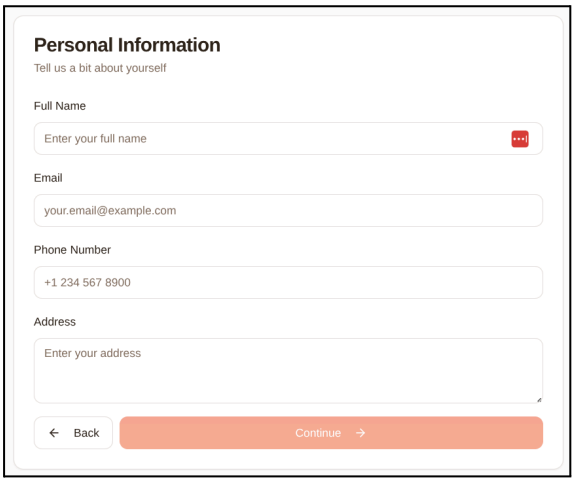
- Landing page
- Virtual Pitch Deck
- Demo frontend prototype with six main components geared towards the various user roles:
    - New user signup flow
    - Consumer Dashboard
    - Payment Agent Dashboard
    - Minigrid Operator Dashboard
    - Investor Dashboard
    - SaaS Operator Dashboard
- Cardano Hydra Implementation
- Technical documents:
    - SolarVillage - EMURGO - Technical Architecture Document
    - SolarVillage - EMURGO - Prototype (PoC) Report
- Other content
    - Catalyst Application video

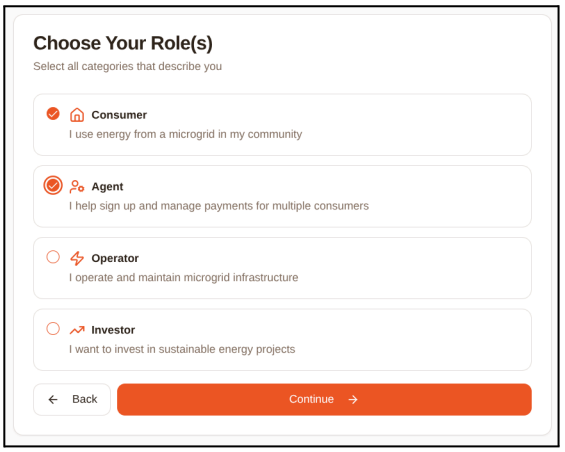
- [Github Repository](#) (access provided on request)

Demo frontend prototype example visuals

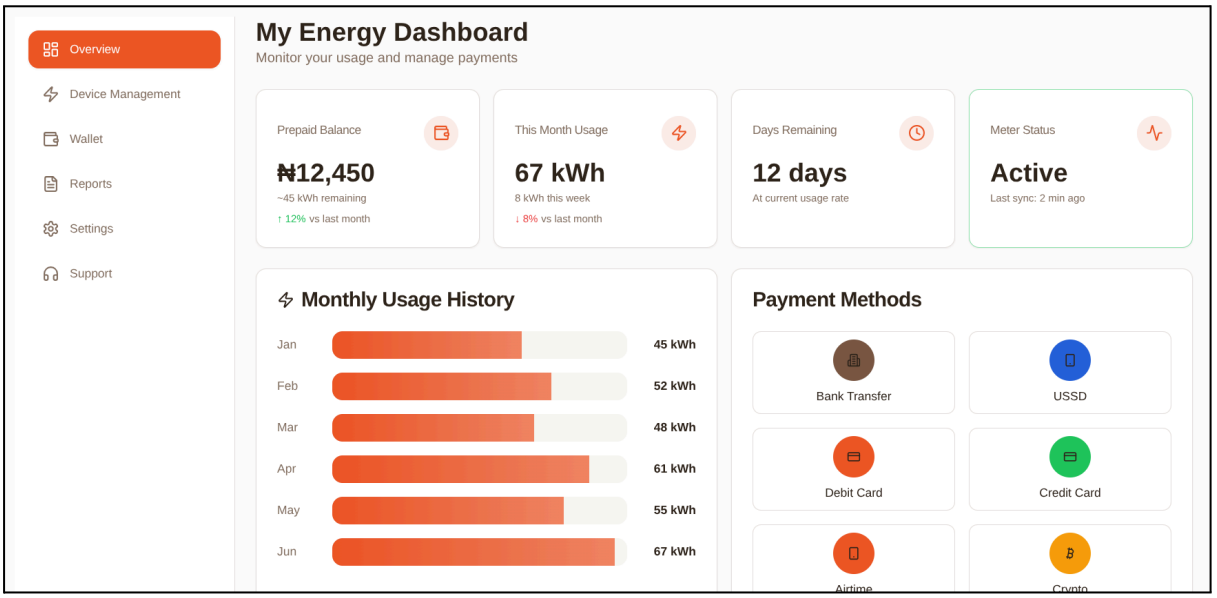
New User Signup Flow



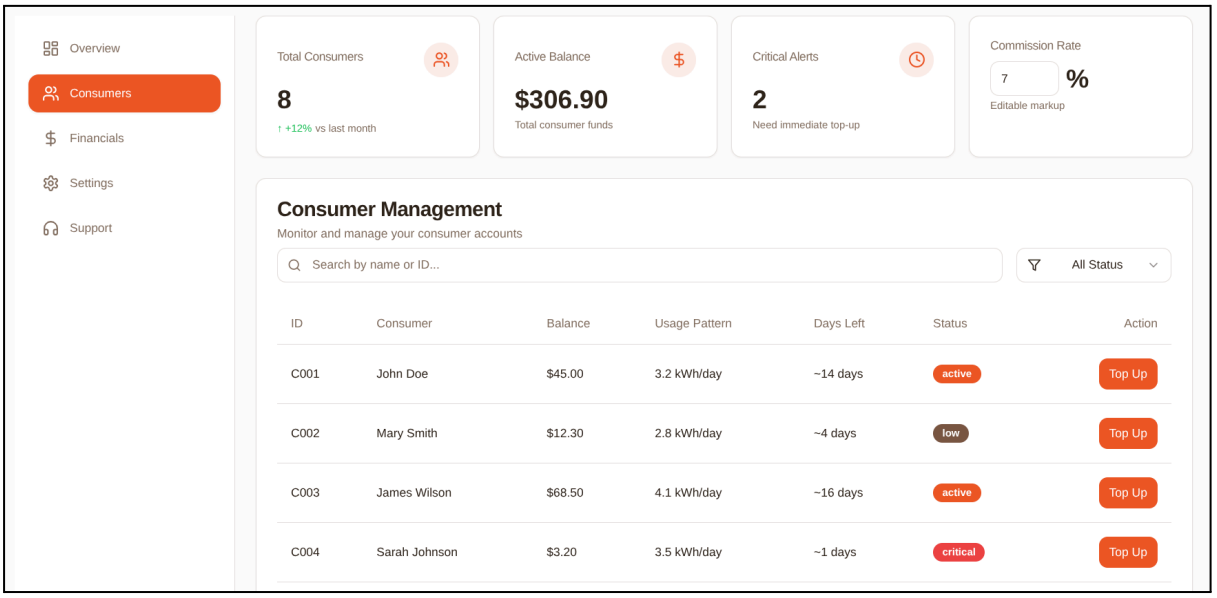




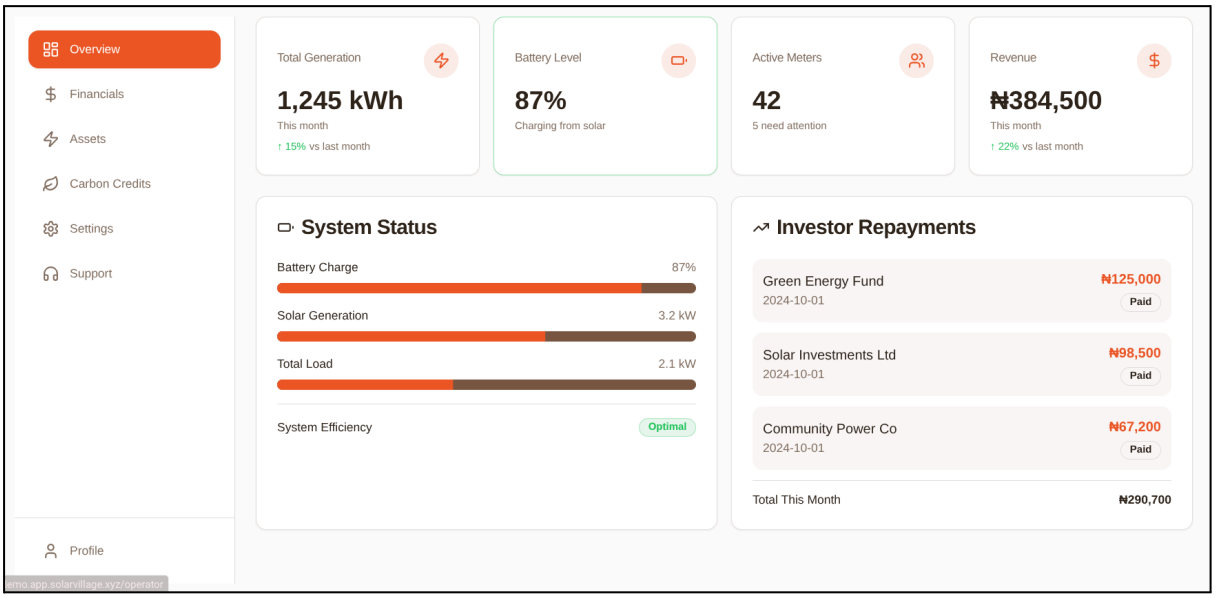
Consumer Dashboard



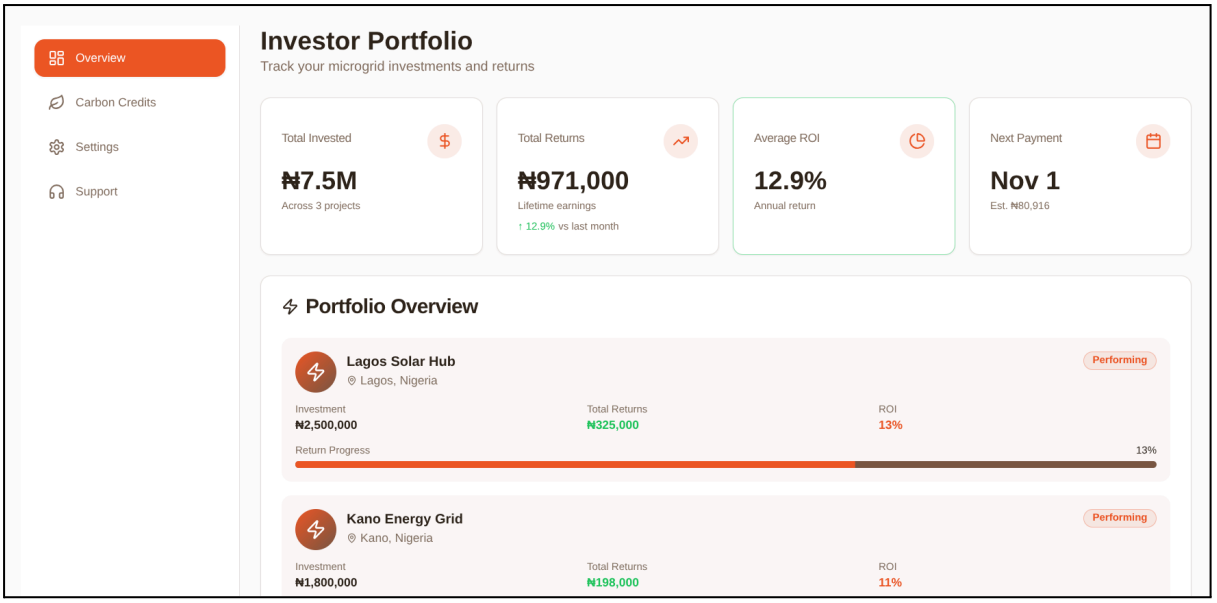
Payment Agent Dashboard:



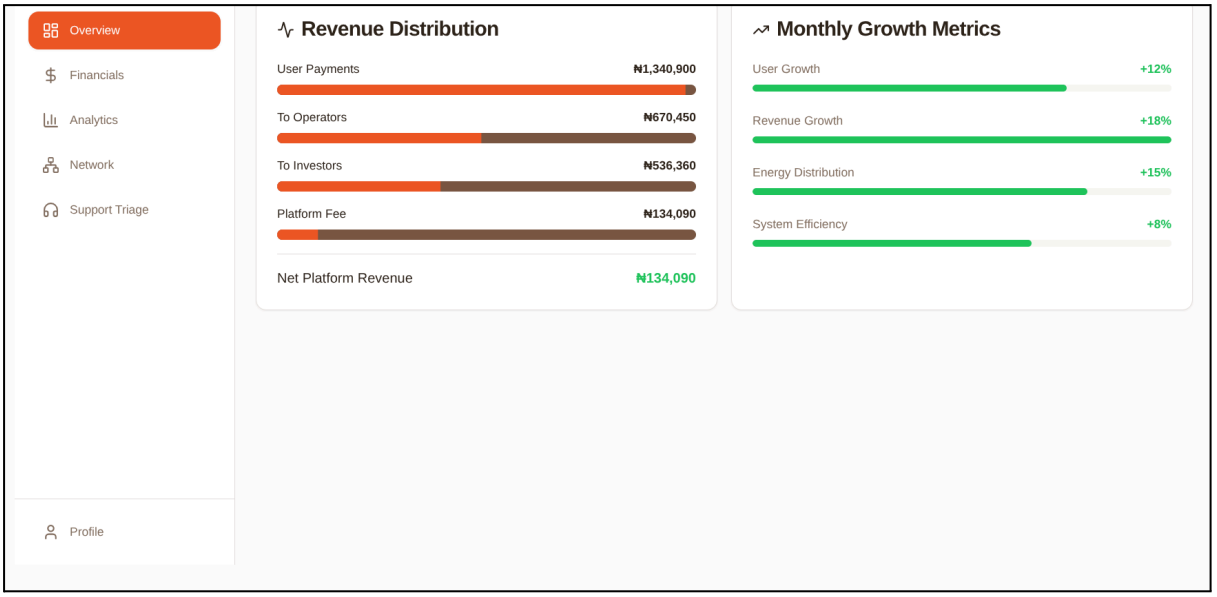
Minigrid Operator Dashboard:



Investor Dashboard:



SaaS Operator Dashboard:

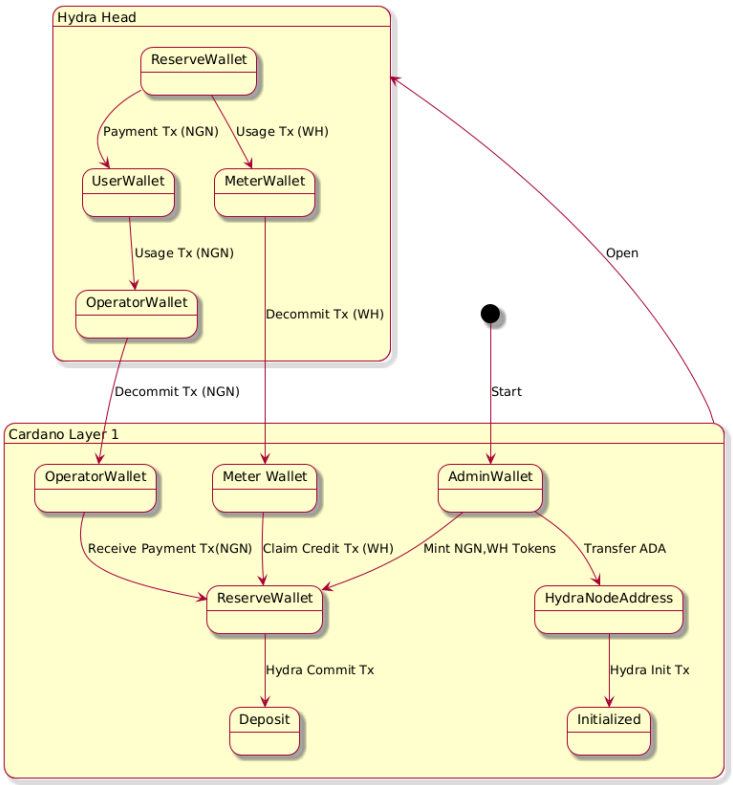


Cardano Hydra Implementation

Below is a high-level summary of the blockchain architecture we've designed for Solar Villge.

Layer	Purpose	Key Technologies
Metering	Collect energy readings every 15-minutes from smart meters.	IoT sensors + MQTT broker + digital signatures
Hydra Heads	Each microgrid operates its own Hydra head for local throughput.	`hydra-node`, Haskell SDK
Anchoring	Daily Merkle root of head data posted to Cardano L1.	Cardano mainnet tx with proof payload
Tariff & Settlement Contracts	Tokenize energy (1 kWh = 1000 Wh tokens) and handle local settlements.	Plutus V3 / Aiken contracts
Carbon Layer	Issue carbon credit tokens only after verified energy data; supports multiple standards.	CIP-68 multi-asset schema
Identity	Device + operator attestation.	Cubid DID API, ClearPass VCs
Applications	Five role-based dashboards: Consumers, Agents, Investors, Grid Operators, Platform Admins.	Next.js frontend + Supabase backend
Audit Interface	Public dashboards for regulators + community.	Grafana / React data visuals

Data flow: Meter → Signed telemetry → Hydra log → Merkle root → Cardano L1 tx → carbon issuance token.



Carbon methodologies: Conversion factors follow Gold Standard & Verra guidelines (kg CO_2/kWh) and Nigeria's forthcoming NCCC registry interface.

Deployment environment:

- Test phase on **Hydra AWS Cloud Instance**.
- Public pilot anchoring to **Cardano pre-prod network**.
- Production anchors on **mainnet** once verified.

5.2 Testing & Feedback Documentation

Summary of user testing activities, challenges, and feedback.

User testing was conducted through a series of structured feedback sessions involving **project partners** and **potential future users**, including practitioners with experience in facilities management and energy operations. These sessions focused on usability, clarity of value proposition, and operational realism rather than abstract feature requests.

Overall reception of the solution was positive. Participants understood the core concept quickly and responded favorably to the combination of transparent energy records, simplified

payment flows, and carbon credit readiness. The dual focus on operator viability and household accessibility was consistently highlighted as a strength.

Key feedback themes included:

- The need for a **more streamlined and frictionless sign-up process**, particularly for first-time users
- Support for **additional payment methods**, especially debit cards, to align with common consumer behavior
- Clearer articulation of **economic value**, including:
    - Cost savings for mini-grid operators
    - Reduced electricity costs or improved price predictability for consumers
    - The long-term revenue potential of carbon credits

No critical usability blockers were identified during testing.

5.3 Feedback Integration Summary

Description of what changes were made based on testing and feedback.

Feedback from testing sessions was incorporated directly into the live demo and design specifications. The following changes and refinements were made:

- **Sign-up flow simplification:**
  Reduced steps and clearer progression to minimize onboarding friction, particularly for non-technical users.

- **Expanded payment assumptions:**
  The system design was updated to explicitly support debit cards alongside other local payment methods, reflecting real-world consumer preferences.

- **Stronger carbon credit framing:**
  The demo and supporting materials were adjusted to more clearly emphasize carbon credits as a core value driver, not an abstract future feature.

- **Clearer value signaling:**
  Messaging was refined to highlight:
    - Financial savings and predictability for mini-grid operators
    - Comparative cost advantages versus diesel generators or unreliable grid access
    - Long-term benefits of verifiable energy records for financing and policy alignment

These updates improved clarity, reduced perceived complexity, and strengthened alignment with stakeholder priorities.

5.4 Pilot-Ready Solution

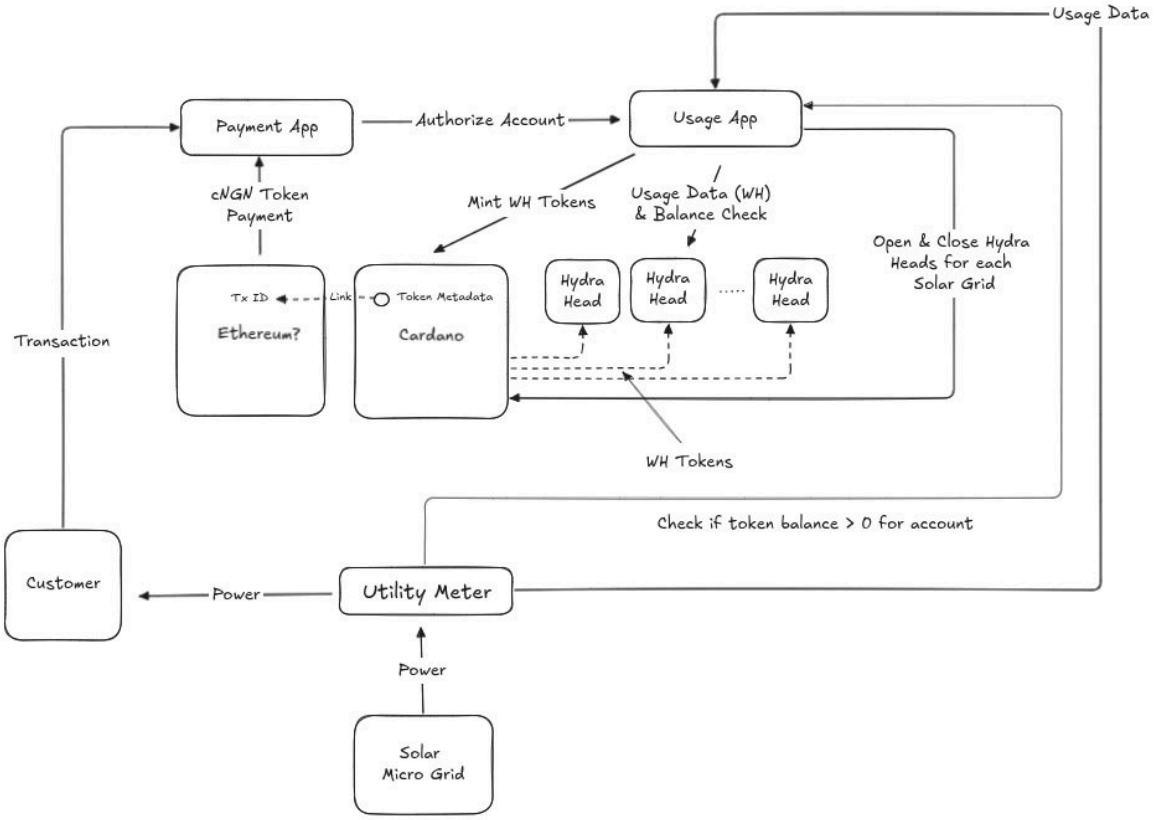
A stable and functional version of the solution, ready for pilot deployment.

At the conclusion of Phase 2, the solution is now **functionally and architecturally ready for pilot build & deployment**, with the following components completed:

- **Blockchain layer:**
  The Cardano Hydra-based architecture for per-mini-grid metering, daily anchoring, and tamper-evident records is fully designed and ready for pilot execution, with on-chain tokens for payment balances and for energy consumption.

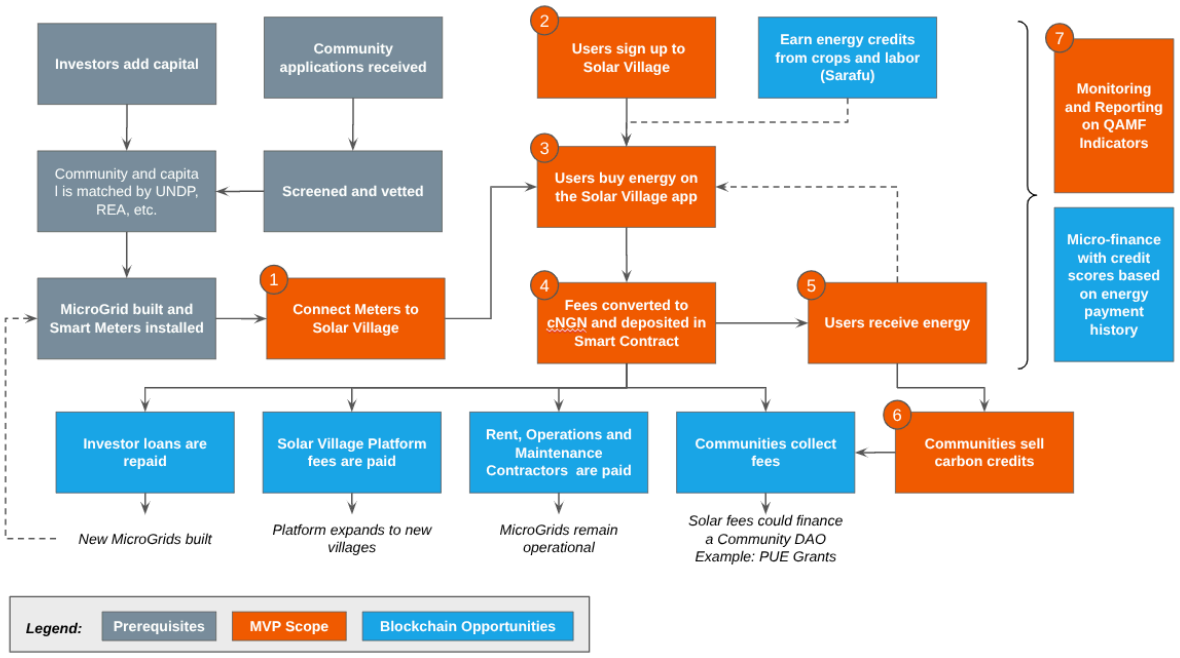
- **Front-end components:**
  Mobile-first web interfaces and core user flows are implemented and demonstrated in the live demo.

- **Operational model:**
  Metering cadence, reporting structure, and oversight workflows are clearly defined and validated with partners.



The scope of the project is clear with 6 process components to be built and piloted during the next phase:

1. Connecting meters to Solar Village
2. User management
3. Payment processing
4. Metering, credit management
5. Provisioning
6. Carbon Credit minting
7. Monitoring & Reporting (QAMF)



The remaining work prior to full pilot launch includes:

- Implementation of the **PostgreSQL backend**
- Any integrations between front end, blockchain components, backend, and metering/payment services, including business logic to support this.

These gaps are well understood, scoped, and do not require changes to the core design assumptions.

More details

For more details, refer to these documents:

- SolarVillage - EMURGO - Technical Architecture Document
- SolarVillage - EMURGO - Prototype (PoC) Report

5.5 Confirmed Pilot Partnerships

Documented confirmation of at least one partner committed to implementation.

The project has confirmed implementation-aligned partnerships with the following organizations:

African Minigrid Program

- Site: https://africaminigrids.org/
- *Partnership nature:* AMP is contributing significant time and energy to Solar Village. One of our core team members since inception is also with AMP. Once Solar Village has been proven in Nigeria, AMP will help with pan-African expansion.
- *Partnership evidence:* xxx

REAN

- Site: https://rean.org.ng/
- *Partnership nature:* We are partnering with REAN to promote our survey and participate in our research study related to the market need for public and verifiable voluntary carbon credits.
- *Partnership evidence:* https://github.com/Solar-Village/hydra-poc/blob/05539f94e661e672d78d907abd5811ec43b86376/docs/REAN_MoU_confirmation.pdf

M3tering Protocol

- Site: https://m3ter.ing/
- *Partnership nature:* We build our tech on open source software / digital public goods from M3tering Protocol, the most modern and trustworthy metering software imaginable.
- *Partnership evidence:* https://github.com/Solar-Village/hydra-poc/blob/05539f94e661e672d78d907abd5811ec43b86376/docs/christwin_MoU_confirmation.pdf

Switch Electric

- Site: https://www.whynotswitch.com/
- *Partnership nature:* Switch provides metering hardware designed to publish verifiable proofs about energy consumption, for undeniable record-keeping and fully traceable carbon credits.
- *Partnership evidence:* https://github.com/Solar-Village/hydra-poc/blob/05539f94e661e672d78d907abd5811ec43b86376/docs/christwin_MoU_confirmation.pdf

Greenpill Network

- Site: https://greenpill.network/
- *Partnership nature:* Two of our core team members are also active Greenpill chapter leads: Greenpill Naija (Nigeria) and Greenpill Toronto (Canada). As a network of dreamers, hackers and doers, Greenpillers help with any blockchain-related aspects of the project.
- *Partnership evidence:* https://github.com/Solar-Village/hydra-poc/blob/05539f94e661e672d78d907abd5811ec43b86376/docs/amio_MoU_confirmation.pdf

Tech And Sun

- *Partnership nature:* T.A.S are electrical engineers with one containerized minigrid already running, and in process of building new microgrids next to universities in Nigeria. They have partnered with us as volunteers to be early pilot installers and beta testers.
- *Partnership evidence:* https://github.com/Solar-Village/hydra-poc/blob/05539f94e661e672d78d907abd5811ec43b86376/docs/amio_MoU_confirmation.pdf

Rural Electrification Agency (REA)

- Site: https://rea.gov.ng/
- *Partnership nature:* REA is overseeing hundreds of minigrid deployments throughout, and have identified 23 of these to be part of the Solar Village pilot implementation.

National Council on Climate Change (NCCC)

- Site: https://climatecouncil.gov.ng/
- *Partnership nature:* NCCC is developing a new national platform for Carbon Credit. Solar Village is collaborating with the developers of the platform and are committed to issue our carbon credits on the NCCC platform.

Solar Foundation

- Site: https://www.solarfoundation.xyz/
- *Partnership nature:* The Solar Foundation participates with ideation and with making introduction and connections within the solar minigrid industry.

UNDP

- Site: https://www.undp.org/
- *Partnership nature:* The original idea for Solar Village was created within the walls of UNDP and continues to be a UDP Nigeria-run project.
- *Partnership evidence:* Two of our core team members work within UNDP Nigeria and help with architecture, program management and government relations.

CUBID Protocol

- Site: https://www.cubid.me/
- *Partnership nature:* CUBID provides decentralized identity management for our users.
- *Partnership evidence:* UNDP partnered with CUBID to create Solar Village as a hackathon entry. One of our core team members works for CUBID.

Firebelly Advisors

- Site: https://www.firebelly.xyz/
- *Partnership nature:* Firebelly provides developer manpower to develop the Solar Village backend tooling.
- *Partnership evidence:* One of our core team members is the managing director of Firebelly.

These partners provide sector legitimacy, implementation context, and policy alignment, and are committed to supporting pilot execution within Nigeria's rural electrification and climate frameworks.

5.6 Stakeholder Alignment Matrix

Defined roles, responsibilities, and coordination plan with key partners.

We have created a deliberately structured collaboration model below, which balances policy alignment, implementation feasibility, and technical delivery. REA and AMP provide the institutional and programmatic anchors required to situate the pilot within Nigeria's rural electrification landscape, while NCCC ensures early alignment with national carbon market and MRV frameworks. Technical expertise is supplied through specialized partners such as Solar Foundation and M3tering Protocol, who support operational assumptions, metering integrity, and data reliability without assuming execution risk. UNDP Nigeria plays a convening and oversight role, coordinating across stakeholders to ensure transparency, learning, and alignment with development objectives. This layered engagement model allows the pilot to proceed efficiently while preserving flexibility for formalization and scale-up in later phases.

Stakeholder / Partner	Role in the Pilot	Key Responsibilities	Level of Involvement	Engagement Modality	Focal Point / Contact Person
REA (Rural Electrification Agency)	Sector anchor and implementation alignment	Alignment with national rural electrification strategy Identification and validation of mini-grid contexts Coordination with mini-grid operators under REA programmes	Lead / Strategic	Programme collaboration / letter of support	AMP / REA Project Manager
NCCC (National Council on Climate Change)	Carbon market and policy alignment	Guidance on alignment with Nigeria's emerging carbon standards Validation of carbon credit readiness approach Advisory on MRV requirements	Advisory / Strategic	Technical collaboration	NCCC Technical Lead (Carbon Markets)
AMP (African Minigrid Program)	Implementation ecosystem partner	Connection to participating mini-grid operators Operational insights and QAMF alignment Support for pilot site selection and scaling pathways	Lead / Support	Programme partnership	AMP Nigeria Programme Lead
SF (Solar Foundation)	Technical and domain expertise	Advisory on solar mini-grid operations Review of operational assumptions and performance metrics Support on best practices for sustainable deployment	Advisory	Informal technical collaboration	Solar Foundation Technical Advisor
MP (M3tering Protocol)	Metering and data infrastructure partner	Expertise on smart metering and data capture Input on meter integration and data reliability Support for metering standards and interoperability	Support / Technical	Technical collaboration	M3tering Protocol Technical Lead

5.7 Budget Proposal (Pilot)

Provide a budget breakdown for pilot implementation. Include estimated cost, unit, quantity, and funding source.

Budget Line Item	Unit / Description	Quantity	Unit Cost (USD)	Total Cost (USD)	Funding Source
Core Hydra + Cardano Integration	Hydra head setup, Aiken contracts, anchoring logic, automation	1 pilot package	$30,000	$20,000	Pending
Frontend Dashboards & Business Logit	Consumer, operator & oversight dashboards (pilot scope)	1 pilot package	$20,000	$10,000	Pending
Data Pipeline & Carbon Layer	Meter ingestion, data normalization, carbon-readiness logic	1 pilot package	$20,000	$10,000	Pending
Backend & Systems Integration	PostgreSQL backend, APIs, auth, component integration	1 pilot package	$20,000	$10,000	Pending
Testing & Infrastructure	Testnets, monitoring, audit scripts, pilot environments	1 pilot package	$15,000	$10,000	Pending
User Testing & Community Engagement	Field sessions, expert reviews, iteration cycles	5 sessions	$0	$0	Pending
Documentation & Knowledge Transfer	Pilot docs, technical handover, reporting materials	1 package	$4,000	$4,000	Pending
Project Management & Coordination	Partner coordination, REA/NCCC liaison, reporting	10 months	$0	$0	Pending
TOTAL ESTIMATED COST				**$64,000**	

5.8 Impact Measurement Framework

Our impact framework prioritizes **verifiable, pilot-appropriate indicators**, focusing on service delivery, transparency, and readiness for scale, rather than speculative long-term outcomes.

This framework has also been documented in more detail in a separate document here

5.8.1 Contribution to relevant SDGs and indicators

The Solar Village project contributes directly to the following three Sustainable Development Goals and indicators:

- **SDG 7 – Affordable and Clean Energy**
  Indicator 7.1.1: Proportion of population with access to electricity
  Contribution: Increased and more reliable access to electricity through solar mini-grids with verified delivery records.

- **SDG 9 – Industry, Innovation and Infrastructure**
  Indicator 9.1.1: Proportion of rural population with access to basic infrastructure
  Contribution: Deployment of digital public infrastructure for operating, monitoring, and financing rural energy systems.

- **SDG 13 – Climate Action**
  Indicator 13.2.2: Total greenhouse gas emissions per year
  Contribution: Measurable displacement of diesel and kerosene usage, with energy data prepared for carbon credit verification.

To translate these SDG-level contributions into actionable and measurable pilot outcomes, the following Key Performance Indicators (KPIs) define how progress will be tracked, assessed, and validated throughout implementation.

5.8.2 Key Performance Indicators (KPIs)

The following table outlines the core Key Performance Indicators (KPIs) used to assess pilot performance and impact. The KPIs are deliberately limited to metrics that can be **objectively measured during the pilot phase**, using system-generated data and structured feedback. Together, they capture service delivery, system reliability, financial transparency, and readiness for future carbon credit verification, providing a balanced view of both operational and development outcomes.

KPI	Measurement Method	Target / Threshold	Notes
Households Served	Registered meters with active usage	≥ pilot baseline (site-specific)	Disaggregated by village
Electricity Delivered (kWh)	Meter data aggregated per grid	Baseline established + monthly growth	Recorded in 15-min intervals
System Uptime	Meter availability logs	≥ 95% uptime	Excluding force majeure
Successful Payment Rate	Confirmed payments vs attempts	≥ 90%	Across all payment channels
Activation Latency	Time from payment → meter activation	< 5 minutes (online)	Codes may vary
Operator Revenue Visibility	On-chain + dashboard completeness	100% of pilot transactions	Audit-ready
Carbon Readiness Coverage	% of energy data meeting MRV standards	≥ 90%	Not issuance, readiness
User Satisfaction	Structured feedback sessions	Majority positive (>70%)	Qualitative + simple scoring
Reporting Completeness	Dashboards populated as designed	100%	UNDP-facing views

5.8.3 Monitoring & Reporting Plan

a) *Frequency:*
b) *Responsible Parties:*
c) *Format:*
d) *Tools:*

This plan describes how pilot performance and impact data will be monitored, reviewed, and reported throughout implementation. The monitoring and reporting plan is designed to support **continuous oversight, learning, and course correction**, while also producing clear, auditable records suitable for UNDP reporting, partner review, and future scale-up decision-making.

Frequency:

- Operational monitoring: Continuous
- Internal reviews: Monthly
- Pilot reporting: Quarterly

Responsible Parties:

- Solution Maker: System monitoring, data aggregation
- Mini-grid operators: Meter data validation
- UNDP Nigeria: Oversight, review, and learning documentation

Format:

- Live dashboards (operational and impact views)
- Monthly internal summary notes
- Quarterly pilot progress reports

Tools:

- Solar Village dashboards
- Cardano / Hydra transaction logs
- Metering system APIs
- Structured feedback notes and session summaries

5.8.4 Risks Assessment & Mitigation Plan

The risks and mitigation measures below reflect both technical and non-technical considerations identified during solution design and testing. The assessment focuses on **pilot-relevant risks** that could affect delivery, adoption, compliance, or partner coordination, and pairs each risk with a practical mitigation approach appropriate to a controlled pilot environment.

Risks	Mitigation
Low user adoption	Community onboarding, simplified flows, multiple payment methods
Technical integration delays	Modular rollout, staged integrations, fallback options
Meter data inconsistencies	Validation rules, operator review, redundancy
Payment failures	Multiple payment rails, retry logic, offline options
Regulatory uncertainty	Early engagement with REA and NCCC, audit-ready design

Risks	Mitigation
Carbon methodology delays	Parallel alignment with multiple standards; readiness focus
Partner coordination challenges	Clear roles, regular check-ins, UNDP convening
Over-reliance on donor funding	Emphasis on operator viability and future revenue streams

5.9 Post-Acceleration Roadmap & Sustainability Plan

5.9.1 Phased Implementation Timeline / Milestones

Short Term (0–6 months)	Medium Term (6–18 months)	Long Term (18+ months)
Focus: **Pilot execution and validation** ● Complete backend implementation and system integrations ● Deploy Cardano Hydra heads for pilot mini-grids ● Launch pilot dashboards for operators and UNDP oversight ● Execute pilot with confirmed partners (REA, AMP, NCCC, etc.) ● Collect operational, financial, and user feedback data	Focus: **Scale readiness and institutional alignment** ● Expand to additional mini-grids within existing partner programs ● Formalize carbon MRV alignment with national and international standards ● Introduce structured pricing for operators (pilot-to-commercial transition) ● Strengthen reporting and audit tooling ● Onboard additional technical and financial	Focus: **Sustainable scale and ecosystem integration** ● National or multi-region rollout in collaboration with government programmes ● Launch carbon credit issuance workflows where approved ● Offer the platform as a managed SaaS or public digital infrastructure service ● Integrate with climate finance, microfinance, or results-based

• Produce pilot evaluation and learning report	partners • Prepare regulatory and procurement documentation for broader adoption	financing mechanisms • Replicate model in other Sub-Saharan African contexts

5.9.2 Budget for Phased Implementation (post-acceleration)

Post-acceleration funding requirements are expected to cover:

- Expansion of infrastructure and hosting
- Additional Hydra head deployments
- Ongoing development and maintenance
- Compliance, audits, and reporting
- Programme coordination and support

Indicative annual budget (post-pilot, directional):

- **$250k–$400k/year** for regional expansion
- **$500k+/year** for pan-African-scale operations

Final budgets will be refined based on pilot results, adoption rates, and regulatory requirements.

5.9.3 Financial Sustainability Model

How will the solution generate income post-pilot? Pricing model (e.g. subscriptions, licensing, transaction fees) and when it kicks in.

Post-pilot, the solution transitions from grant-supported deployment to a **mixed sustainability model**:

- **Subscription fees** paid by mini-grid operators for platform access, reporting, and compliance tooling

- **Transaction or usage-based fees** tied to metered energy volume or reporting services

- **Carbon-credit-related revenues**, once issuance is enabled, where platform services support MRV and verification

- **Institutional licensing or service agreements** with governments or development programmes

Revenue generation is expected to begin **after pilot validation**, with gradual ramp-up during the medium-term phase to avoid burdening early adopters.

5.9.4 Partnership and Co-Financing Commitments

Which partners will be supporting the project (technical, financial, institutional) and what would be their contributions?

The project benefits from a layered partnership model:

- **Institutional partners (REA, AMP, NCCC):** policy alignment, implementation context, programme integration

- **Technical partners (M3tering Protocol, Solar Foundation):** metering, data integrity, domain expertise

- **UNDP Nigeria:** convening, oversight, learning, and credibility

Financial contributions post-pilot are expected through:

- Blended finance structures (grants + concessional capital)
- In-kind technical and operational support
- Future service agreements with implementing agencies

Formal co-financing commitments will be pursued following successful pilot evaluation.

5.9.5 Fundraising Plan

Plan for securing funding through grants, ecosystem funds, investors or other sources. Targeted sources and fundraising timeline.

The fundraising strategy follows a **sequenced approach**:

- **Short term:**

    - UNDP innovation or scale-up facilities

    - Climate and energy-focused grants

- **Medium term:**

    - Climate finance facilities
    - Development banks and ecosystem funds
    - Results-based financing mechanisms

- **Long term:**

    - Impact investors aligned with energy and climate outcomes
    - Carbon market participants
    - Public–private partnership models

Fundraising activities will be anchored to **verifiable pilot results**, reducing perceived risk for funders.

5.11 Legal & Compliance Considerations

Summary of key legal, regulatory, or compliance considerations.

Key legal and compliance considerations include:

- Alignment with Nigeria's energy regulation framework
    - Develop solution in close partnership with the Rural Electrification Agency (REA)
- Compliance with emerging national carbon market regulations
    - Develop solution in close partnership with the National Council on Climate Change (NCCC)
- Data protection and privacy considerations for household and operator data
    - Our solution maker CUBID Protocol is a leading authority on decentralized identity management and privacy-preserving technology
- Financial compliance related to payment processing and settlement layers
    - Develop payment solution in collaboration with the Africa Stablecoin Consortium and Securities and Exchange Commission of Nigeria (SEC) to use Compliant Naira (cNGN) issued by Wrapped CBDC Lt
- Smart contract and blockchain governance best practices
    - Develop the blockchain solution in close partnership with EMURGO Labs, Cardano Foundation, and faculty from George Brown Polytechnic

The solution is intentionally designed as **infrastructure and record-keeping**, minimizing regulatory exposure while enabling future compliance-aligned use cases.

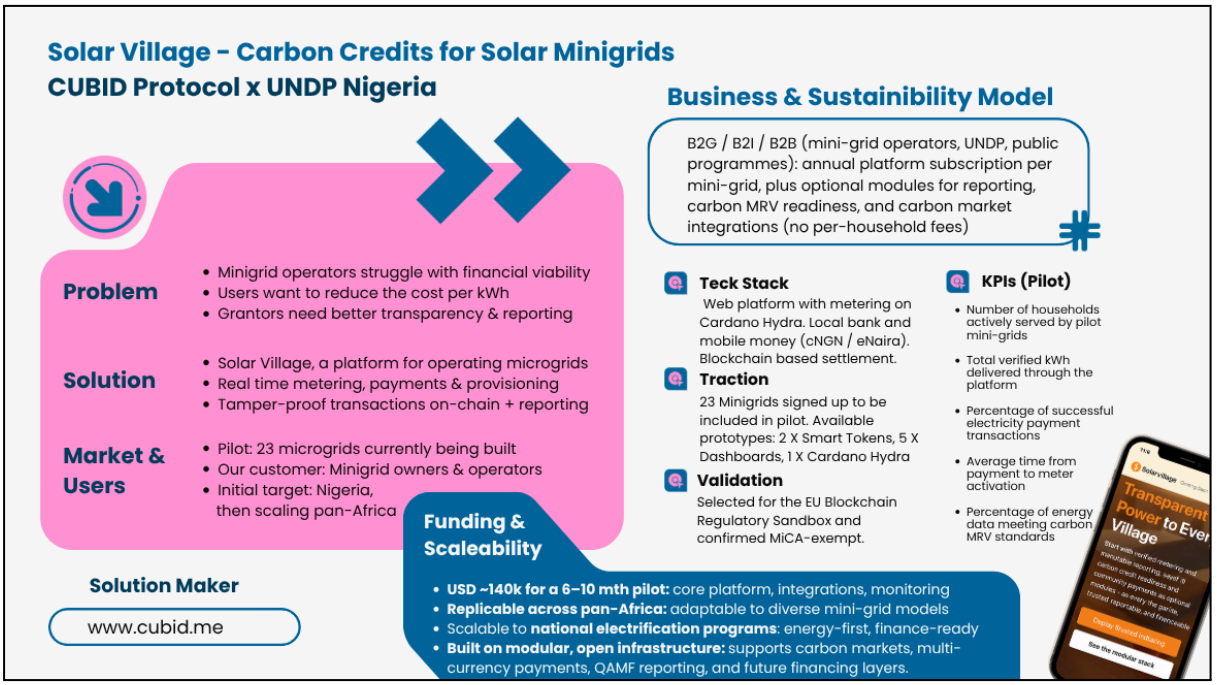
5.12 Visibility Assets / Communication Materials

1–2 slides, visuals, or storylines for use by UNDP or partners to promote your solution.

The following assets are available or in preparation for UNDP and partner communications:

One-page summary graphical

See it on Canva here



One-page summary text document

See it in Google Docs here

Nigeria: Operations & Carbon Credits for Rural Solar Mini-grids

Challenge Owner: UNDP Nigeria

Solution Maker: Cubid Protocol – a decentralised identity and payments protocol that helps apps verify real users and anchor tamper-evident transactions on-chain.

System-level Challenge

Many rural communities in Nigeria are unlikely to connect to the national grid soon and still rely on expensive, unreliable generators. Solar minigrids are a viable alternative, especially in farming areas where energy can power irrigation, milling and cooling, but their long term viability depends on predictable tariff collection and clear financial records. Investors, governments and grantors struggle with transparent reporting from projects, minigrid operators struggle with financial viability and users want to reduce the cost per kWh.

What we are Piloting

The initiative introduces **Solar Village**, a platform for operating microgrids. We build digital public infrastructure for metering that lets operators connect their minigrid and smart meters and add several value-add services:

1. Real time reporting, satisfying the need of investors, governments and grantors
2. Carbon credits can be "minted", verified and sold on the open market
3. Post-paid purchases. Households buy electricity with debit cards or in the same way they buy mobile airtime.

This strengthens the business case for minigrids, creates verifiable on-chain histories for households and operators, and opens a path to future microfinance for productive-use equipment.

How It Works

Metering data is recorded every 15 minutes and stored transparently on a blockchain. Each minigrid is registered in a digital asset inventory linked to a reference on the blockchain. This means every kWh has a consumption record that can be tied back to specific equipment and connections.

Carbon Credits can be minted with regular frequency and sold to generate supplemental income for operators, offsetting their costs. Operators and UNDP teams use a shared dashboard to view electricity production, technical performance, etc. Over time, the same rails can support credit for farming equipment and other productive uses that depend on reliable power.

Households in selected villages can purchase prepaid electricity through local payment channels and confirm top ups. Once payment is confirmed, Solar Village activates the household meter either through an online interface or time limited codes, so any user can participate regardless of device.

Snapshot

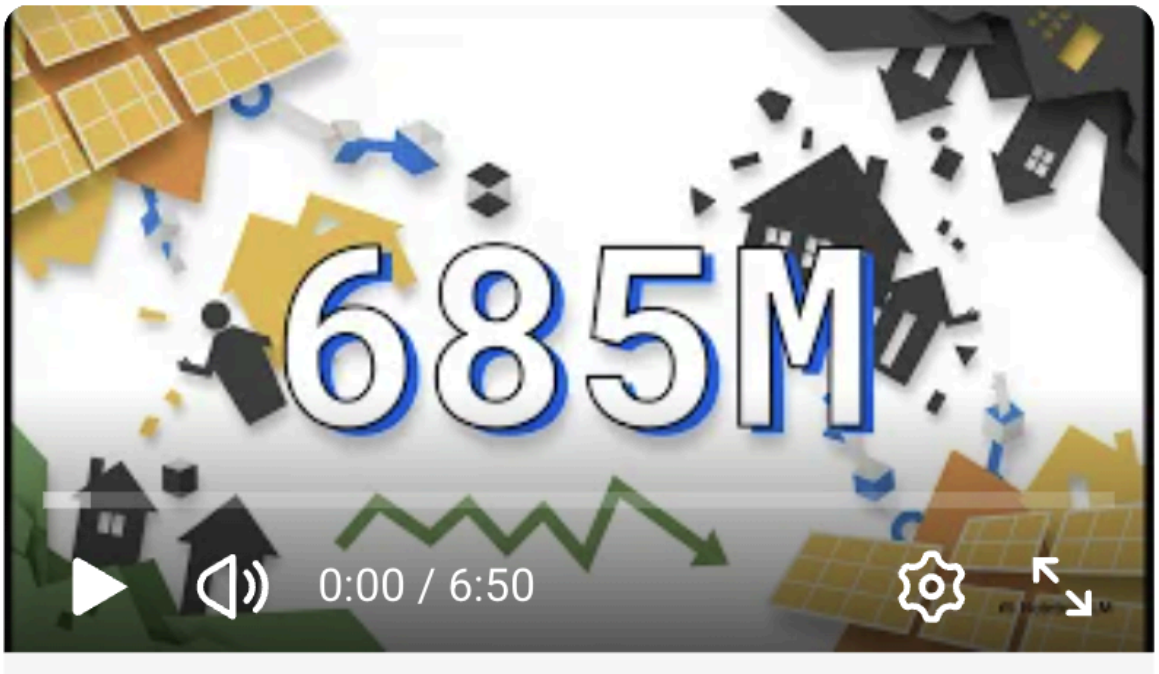
Key users: Rural households, solar minigrid operators, community cooperatives, local agricultural producers and micro-entrepreneurs.

Tech stack: Solar Village platform with Cardano Hydra back end and React + Typescript mobile web front end. Local bank and mobile money integrations and blockchain based settlement in eNaira.

Scale: Designed for replication across Nigeria and adaptable to rural electrification initiatives in other parts of Sub-Saharan Africa.

SDGs: 7; 8; 9; 11

5 min intro video



https://youtu.be/iw7Ia9X2uNc

Other

- Visuals used in this document and elsewhere are in Google Slides here
- Architecture diagrams
- Clickable demo of the Solar Village platform here

- Short pitch narrative for donor and investor audiences here
- Longer benefits video narrative for general audiences here

5.13 Pitch Deck

Concise and compelling presentation for stakeholders, funders, and partners. (Please upload to your google drive folder)

We have three versions of the pitch deck:

- A dynamic pitch deck is incorporated into the website, find it as an online pitch here
- A Google Slides version of main pitch deck is here with a video version here
- A Canva version of a pitch deck is being developed here

5.14 Final Demo Presentation

An engaging walkthrough of the live solution and its implementation roadmap at the end of the program.

Video on youtube: https://youtu.be/xqn6T30fti0

Longer video from the demo meeting - need to get link from Burcu who recorded it.

We also have a demo presentation made for Catalyst here.

Deliverable Completion Table

Please indicate your intended completion week (or date) for each deliverable. You can revise this plan as the project evolves.

Deliverable	Target Completion (Week # or Date)

1. Challenge Definition (incl. local context, SDG contributions, partners, impact)	Done
2. User Mapping & Testing Strategy (personas, user stories, feedback loop)	Done
3. Solution Overview (core features, tech stack, blockchain functionality)	Done
5.1 Solution Demo (PoC)	Done
5.2 Testing & Feedback Documentation	Done
5.3 Feedback Integration Summary	Done
5.4 Pilot-Ready Solution	Done
5.5 Confirmed Pilot Partnership(s)	Done
5.6 Stakeholder Alignment Matrix	Done
5.7 Budget Proposal (Pilot)	Done
5.8 Impact Measurement Framework (KPIs, SDG indicators, risk plan)	Done
5.9 Sustainability Plan & Post-Acceleration Roadmap (implementation timeline, budget, partners, fundraising)	Done
5.11 Legal & Compliance Mapping	Done
5.12 Visibility Assets / Communication Materials	Done
5.13 Pitch Deck	Drafting